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# FUNDAMENTALS OF PAYMENTS

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Some explanations of the key terms and key stages in a payment, itself.



## FUNDAMENTALS

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There are some terms that are important to be aware in terms of payments in general, for those not familiar with the industry:

- In any transaction between a consumer and a merchant, the parties involved are usually:
  - The issuing bank that issued the card of the consumer;
  - The acquiring bank where the merchant has an account;
  - The card company (e.g. VISA, Mastercard);

The handling of CHD by the merchant can be performed in a CP (Card Present) (e.g, terminal in a shop) or CNP (Card Not-Present) conditions (e.g., buying online);

- The PCI DSS apply to both, although data vary slightly;



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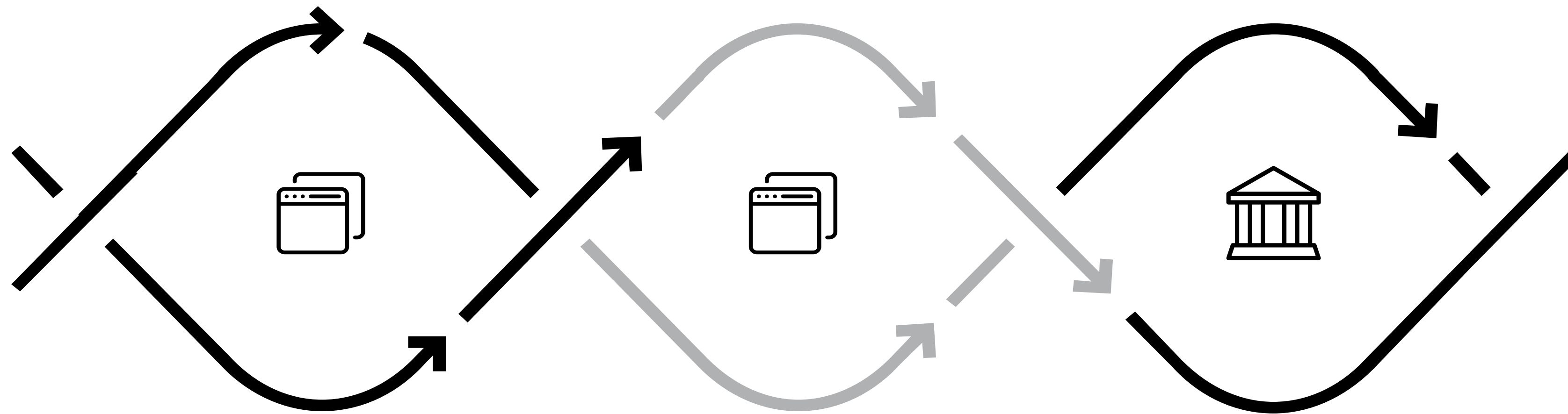
Then, we have the payment flow itself. A payment from a consumer to a merchant, relative to a purchase of products/services, has three major stages:

1. Authentication
  - a. The merchant sends the details to the card company;
2. Authorisation
  - a. The card company obtains authorization from the issuing bank (or not);
3. Clearing and Settlement
  - a. Sometime later, the merchant sends all transactions to their acquirer bank, which through the card company collects the values from all clients' issuing banks;



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### 1. AUTHENTICATION

The merchant sends the details of the transaction to the acquirer bank, which sends them to the card company.

### 2. AUTHORISATION

The card company sends the request to the issuer bank, which authorizes (or not). The customer receives a receipt on the POS.

### 3. CLEARING AND SETTLEMENT

At end of day (or within 24h), the merchant sends all transactions to the acquirer bank, which sends them to the card companies, which sends them to the issuer bank. Issuing banks make transfers to acquiring bank through the card company.



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The first step, Authentication, is all about the merchant communicating the transaction to the card company.

- When you buy something with a VISA card and see, on the POS, “\$19.99, enter PIN”, the merchant will request authorization from their bank to charge \$19.99, which the bank will communicate to VISA;

The second step, Authorisation, is all about the bank of the cardholder accepting the charge.

- When you see on the POS, “Processing”, after entering the PIN, VISA communicates this charge to your issuing bank, which authorizes it (if there’s enough balance);
- Upon authorized, a hold is placed on the \$19.99 + receipt;







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The third step is Clearing and Settlement, two common transaction backoffice operations, within the next 24/48h.

- Clearing means that, at some point, the issuing bank will verify that you have the capital, that you are the account owner, and other simple checks;
- Settlement is the final, legal settlement of ledgers. Your bank account has less \$19.99, the merchant's has plus;

At the end of the day (or 24/48h), the merchant sends all transactions, including your \$19.99, to the acquiring bank. The acquiring bank, in turn, sends them all to the card company.

- The card company charges the issuing bank for the authorized ones and deposits the acquirer bank the values;



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### EXAMPLES

#### **/01 CHARGEBACKS**

When a customer believes they have been tricked or defrauded, the issuing bank performs a “chargeback” - a request to the acquirer bank to return the money.

#### **/02 NETTING AND CLEARING**

The reason why the clearing process takes 24/48h is due to “netting”. Some transactions are +, some are -. These are grouped, and only the net is transacted.

#### **/03 NOT AUTHORIZED**

In the second stage, Authorization, if the customer doesn't have enough balance (or other problems), authorization is rejected, and the process stops there.

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### KEY TAKEAWAYS

#### /01 KEY TERMS

In a transaction, you usually have the merchant and their bank (acquirer), a customer and their bank (card issuer), and a card company at the center of it all.

#### /02 THREE MAJOR STAGES

There are 3 major stages: Authentication (merchant to CC), Authorisation (CC to issuing bank), and Clearing/Settlement (acquiring bank to CC to issuing).

#### /03 CLEARING + SETTLEMENT

Clearing and settlement are two transaction backoffice operations. “Clearing” the transaction makes final checks, and “settling” settles ledgers.